

F. Assignment: Multi-Agent Decentralized Waste Collection and Recycling System

Objective:

Design and implement a decentralized waste collection and recycling management system using **SPADE**. The system will simulate a smart city where multiple agents (waste bins, collection trucks, recycling centers, and citizens) collaborate to optimize waste collection, reduce overflow, and maximize recycling efficiency without relying on a central controller.

Problem Scenario:

Urban waste management is often centralized, leading to inefficiencies such as overflowing bins, unnecessary truck routes, and poor recycling rates. A decentralized, agent-based approach allows bins, trucks, and recycling centers to coordinate autonomously. Agents must share information about bin fill levels, truck capacity, and recycling center availability, while adapting to dynamic conditions such as traffic congestion, sudden waste surges (festivals, events), or equipment breakdowns.

Key Features of the Assignment:

1. Smart Bin Agents:

Each bin monitors its fill level and type of waste (general, recyclable, organic). When nearing capacity, it communicates with nearby trucks.

2. Truck Agents:

Represent waste collection vehicles with limited capacity and fuel. They plan routes dynamically, responding to bin requests and recycling center availability.

3. Recycling Center Agents:

Manage intake capacity for different waste types. They negotiate with trucks to accept deliveries and balance loads across centers.

4. Citizen Agents (Optional):

Generate waste at varying rates, simulating real-world variability in bin fill levels.

5. Resource Management:

Agents must manage limited truck capacity, fuel, recycling center throughput, and time.

6. Decentralized Coordination:

No central waste authority exists. Bins, trucks, and centers negotiate peer-to-peer to optimize collection and recycling.

7. Dynamic Environment:

Events like festivals, traffic jams, or truck breakdowns force agents to adapt their plans.

8. **Collaboration Protocols:**

Use **Contract Net Protocol** for task delegation (e.g., a bin requests collection, trucks bid based on proximity and capacity).

9. **Performance Metrics:**

- Percentage of bins collected before overflow
- Recycling rate achieved
- Average truck fuel/time efficiency
- Responsiveness to dynamic events
- Effectiveness of decentralized collaboration

10. **Agent Specialization:**

- **Bins:** Monitor and request collection
- **Trucks:** Plan and execute routes
- **Recycling Centers:** Manage intake and processing
- **Citizens (optional):** Generate waste dynamically

Suggested Development Phases:

Week 1-2:

- Introduction to MAS and SPADE.
- System design: define agent types, communication protocols, and city environment.
- Initial setup: implement basic bin and truck agents.

Week 3:

- Implement communication between bins and trucks.
- Basic collection logic: trucks respond to bin requests.

Week 4:

- Add resource constraints (truck capacity, fuel, recycling center limits).
- Introduce dynamic events (traffic, surges in waste).
- Agents adapt routes and collection strategies.

Week 5:

- Implement advanced collaboration protocols (Contract Net Protocol).
- Add recycling center agents to balance intake.
- Trucks negotiate with centers for optimal delivery.

Week 6:

- Build a visualization interface showing city layout, bins, trucks, and recycling centers.
- Test under different scenarios (waste surges, traffic jams, equipment failures).
- Evaluate performance with defined metrics.